# Notes from Ben, not part of the talk

- Side chain: now explained on slide 2 in one clause (Ben, 14 October). The whole deck was revised on 14 October along the four facts of slide 6, see `WORK-SPLIT.md`.

# Highly accurate protein structure prediction with AlphaFold

Paper by Jumper et al., *Nature* 596 (2021)

Evita Bosch Pokunevich  Ben Bullinger

Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

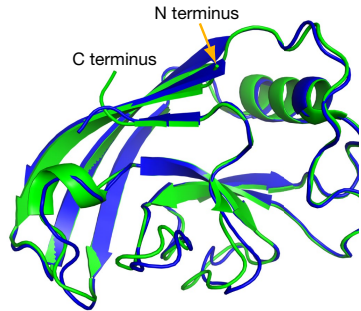
# Outline

1. The problem
2. The benchmark
3. The network
   - The data
   - The network, in detail
4. Training
5. What makes it work
6. Conclusion
7. Questions

# The problem: sequence to structure

SIFSYITESTGTPS...

sequence of amino acids. Each has the same backbone and its own side chain




3D coordinates of all heavy atoms
Prediction in blue, experiment in green

An experiment takes months to years

Billions of sequences are known

**The problem is more than 50 years old**

Sequence and structure are target T1049 of CASP14, the blind prediction contest, Protein Data Bank (PDB) entry 6Y4F, Fig. 1b.

# The benchmark: CASP14

CASP14: Critical Assessment of protein Structure Prediction

May-July 2020

Blind test - experimental structures are not revealed until the end of experiment.

Å (angstrom): a unit of length equal to $10^{-10}$ meters.

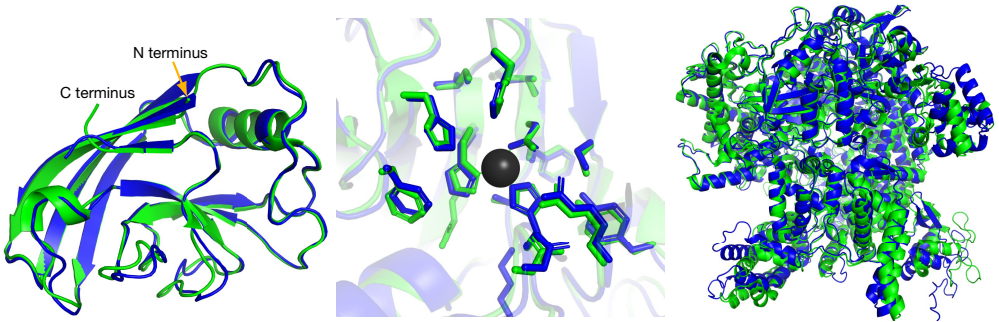
Smaller RMSD (root mean square deviation of the atom positions) means a closer structural match.

**Show the average prediction  hist from the paper**

**The median backbone error is below one atom width**

Redrawn from Fig. 1a. The error is the Cα r.m.s.d.$_{95}$ on 87 domains, with 95% confidence intervals.

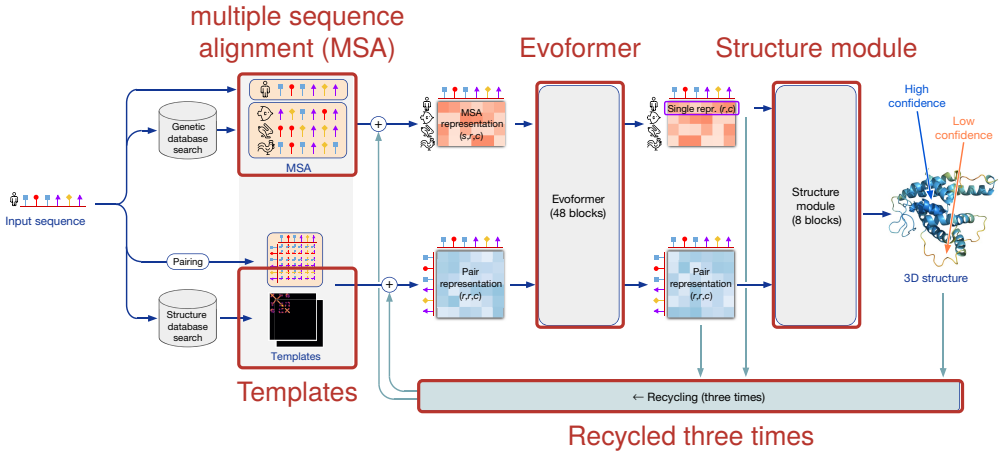
# Prediction and experiment superposed




AlphaFold experiment

**Prediction and experiment overlap, also in the side chains**

# The pipeline: MSA in, coordinates out




**Coordinates come out of the network itself**

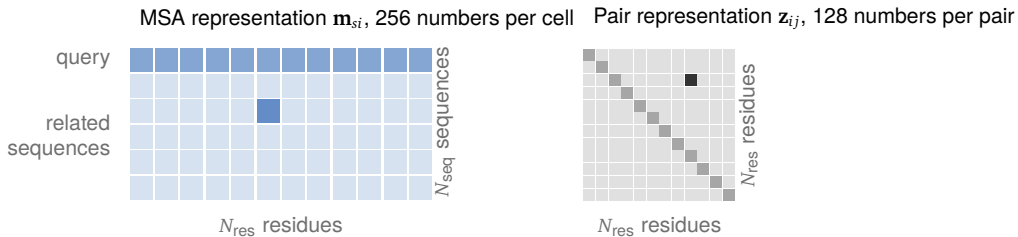
# Four properties, four design choices

| | | Fact | Design choice in AlphaFold 2 |
|---|---|---|---|
| 1 |  | An MSA is a set of sequences. A protein is a chain of residues. | Attention over the sequences themselves, along the rows and the columns. |
| 2 |  | Distances in 3D must close triangles. | The pairs are the edges of a residue graph, updated through triangles. |
| 3 |  | A structure is the same after any rotation or translation. | One frame per residue. Attention uses only distances in the frames. |
| 4 |  | A protein and its mirror image are different molecules. | The loss compares atoms in local frames. A mirror image is penalised. |

**Removing any one of the four costs accuracy. The ablations measure how much.**

# The two representations

## Two tables of numbers, one per MSA cell and one per residue pair

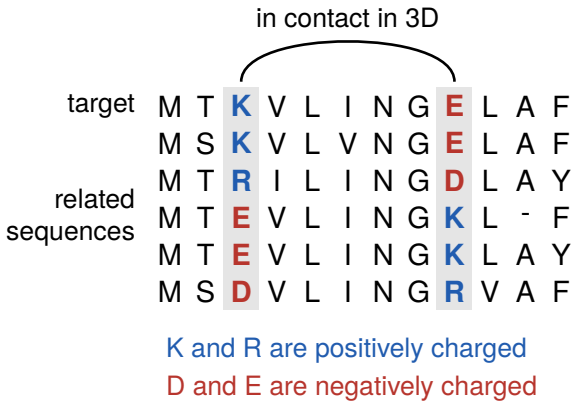



One vector per cell of the alignment, and one per ordered pair of residues

**Everything the network learns is written into these two tables**

Supplementary Information (SI) 1.5, Algorithms 3 and 4. Sizes in training: 128 sequences, crops of 256 residues (SI Table 4).

# The input: MSA representation




A row is a related protein

A column is an aligned position

Contacts may change together

**The network reads the sequences themselves**

D. S. Marks et al., *PLoS ONE* 6, 2011. The alignment is an illustration, not data.

# The input: Pair representation

## What $z_{ij}$ holds and where it comes from

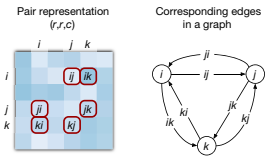



Fig. 3b, the pairs as directed edges

128 numbers for every ordered pair of residues, the edge $i \to j$ of the graph

Started from the two amino acids and their distance along the chain, clipped at 32. Templates and the previous cycle enter as distances.

At the end, the distogram reads a distance distribution for every pair from it

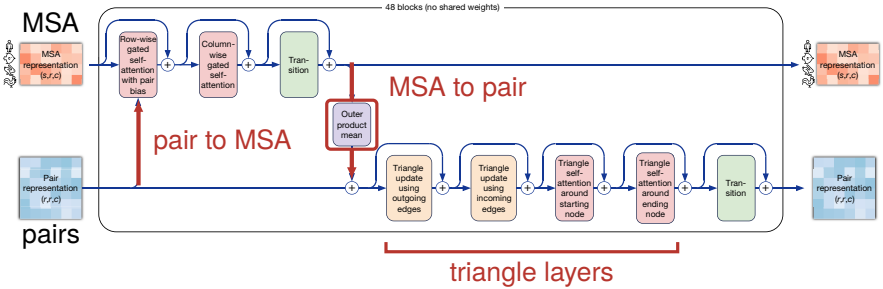
**Where the structure is held before there are coordinates**

SI 1.5, Algorithm 4 (relative position), 1.7.1 (templates), 1.10 (recycling), 1.9.8 (distogram).

# The Evoformer block



**1. The sequences in an MSA are a set. The residues are a chain.**

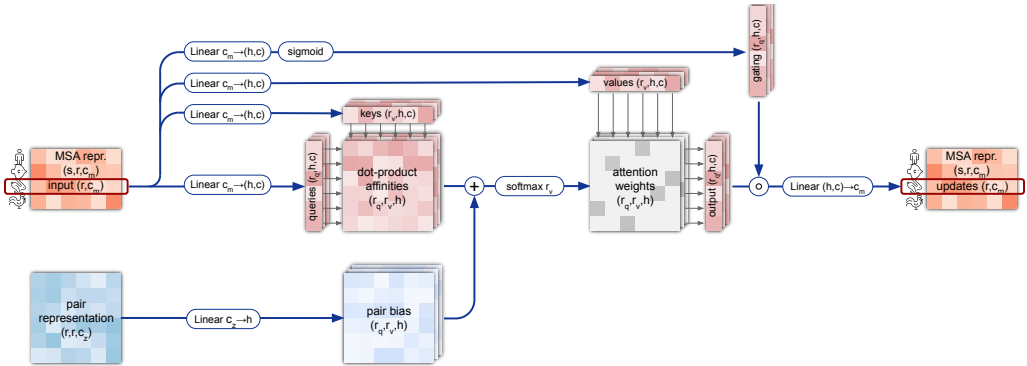



Attention along the rows and the columns. MSA and pair representation exchange information

**The paper calls this graph inference. The edges are residues in proximity.**

# Evoformer: row attention with pair bias

**Supplementary Fig. 2, Algorithm 7**

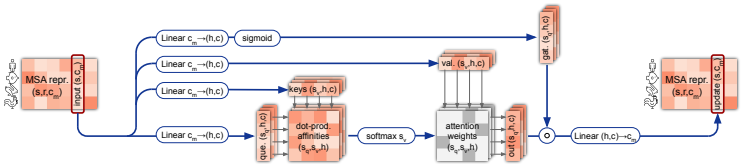



One sequence at a time: residue $i$ attends to every residue $j$ of the same sequence

**The pair representation is added to the logits. The pairs steer the MSA.**

# Evoformer: columns and outer product mean

**Supplementary Figs. 3 to 5, Algorithms 8 to 10**




One residue at a time: the sequences attend to each other at position $i$, then a two-layer perceptron per cell

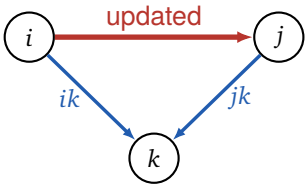
Outer product mean: the outer product of columns $i$ and $j$, averaged over the sequences, is added to $\mathbf{z}_{ij}$

**Here the alignment writes into the pairs, in every one of the 48 blocks**

# Evoformer: triangle updates

## 2. Distances in 3D must close triangles.

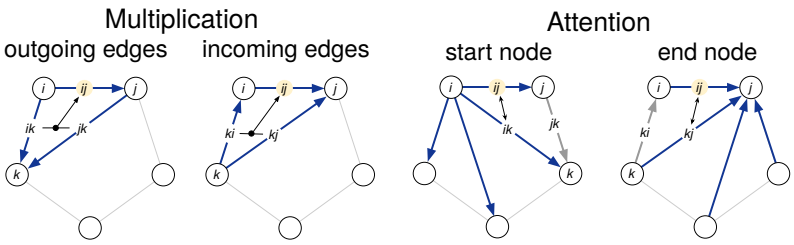





The pair representation is the edge set of the graph (Fig. 3b)

Edge $ij$ is updated from the edges $ik$ and $jk$

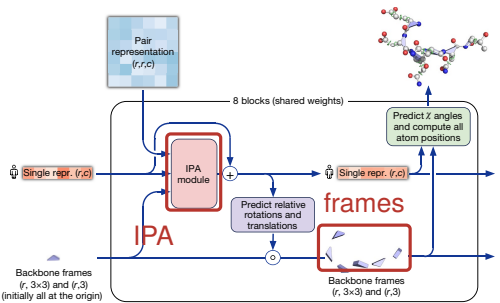
**The triangle inequality acts as an inductive bias**

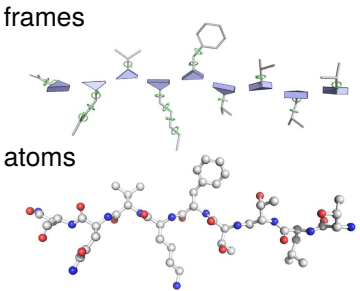


# Structure module: residues as rigid bodies



### 3. A structure is the same after any rotation or translation.




A frame is one rotation and one translation per residue

Invariant point attention (IPA) updates the residues, then all frames move at once

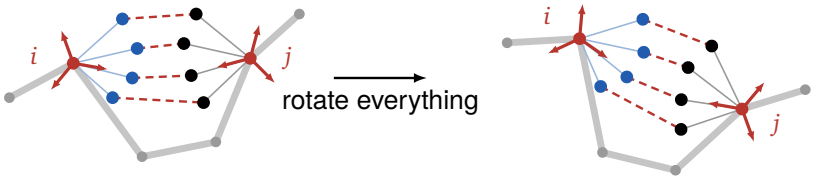
**Nothing forces neighbouring residues to stay connected**

# Invariant point attention



### 3. A structure is the same after any rotation or translation.

$$a_{ij} = \text{softmax}_j\Big( \underbrace{\boldsymbol{q}_i^\top \boldsymbol{k}_j}_{\text{usual attention}} + \underbrace{b_{ij}}_{\text{pair bias}} - \underbrace{\textstyle\sum_p \left\| T_i \circ \vec{\boldsymbol{q}}_{ip} - T_j \circ \vec{\boldsymbol{k}}_{jp} \right\|^2}_{\text{distances between 3D points}} \Big)$$




Points are placed in the frames of $i$ and $j$

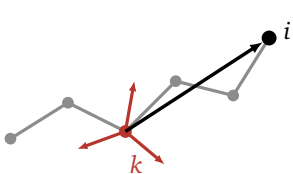
The distances stay the same

**The weights are invariant to global rotation and translation**

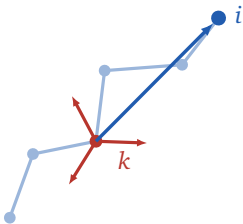
The formula is Algorithm 22 of the supplement for one head, without scaling constants.

# Frame aligned point error (FAPE)

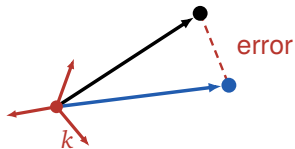
4. A protein and its mirror image are different molecules.




experiment




prediction




both, seen from frame $k$

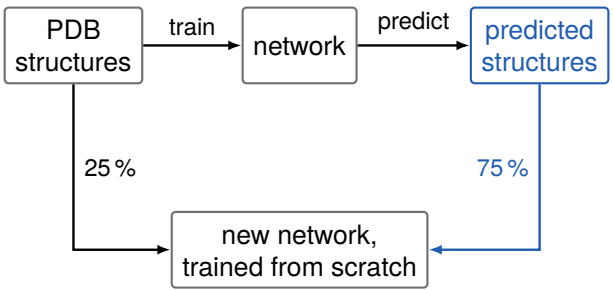
$$\text{FAPE} = \frac{1}{Z} \underset{k,i}{\text{mean}} \min\left(d_{\text{clamp}}, \left\|(\boldsymbol{R}_k, \boldsymbol{t}_k)^{-1}\boldsymbol{x}_i - (\boldsymbol{R}_k^{\text{true}}, \boldsymbol{t}_k^{\text{true}})^{-1}\boldsymbol{x}_i^{\text{true}}\right\|\right)$$

**A rotation changes nothing. A mirror image is penalised.**

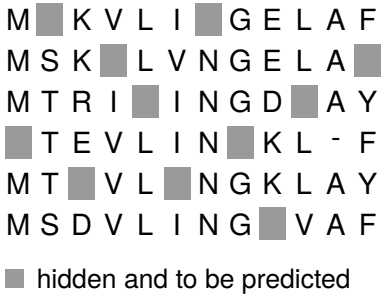
Supplementary Information, Algorithm 28, with $Z = d_{\text{clamp}} = 10\,\text{Å}$. The drawing is an illustration.

# Training data: PDB and self-distillation

Self-distillation adds 350,000 structures



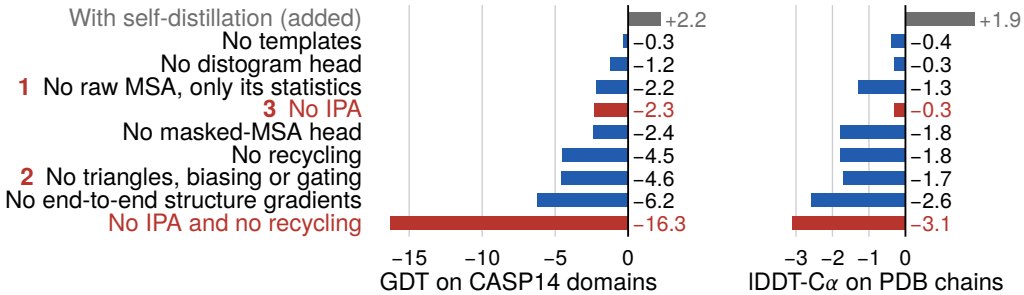

A masked-MSA loss is added




**Training uses sequences without structures**

Self-distillation follows Q. Xie et al., 2020. Masking follows BERT, J. Devlin et al., 2019.

# Ablations: the cost of each component

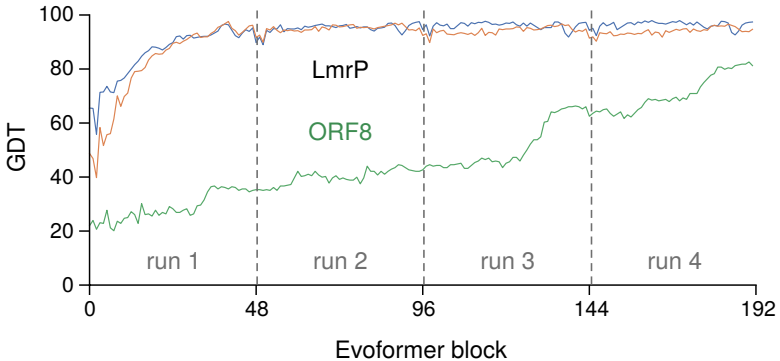



4 With a mirror-blind loss instead of FAPE, the model often builds the mirror image

**Each of the four choices costs accuracy on CASP14 when it is removed**

Redrawn from Fig. 4a, the mean change against the baseline over the training runs of a row. GDT is the global distance test, lDDT the local distance difference test, both 0 to 100. Line 4: Suppl. Fig. 9.

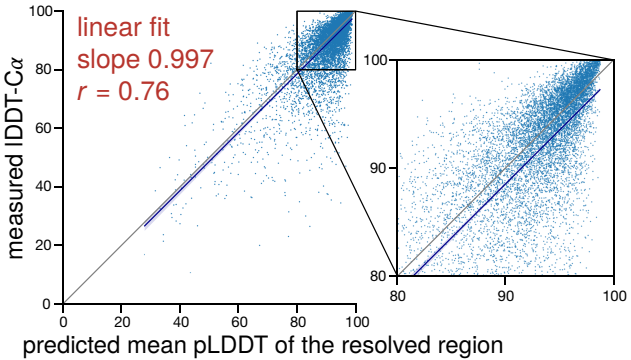
# Intermediate structures in the trunk




The trunk is frozen and every block gets a structure module

**Some proteins are solved early and others late**

# Confidence: pLDDT against measured lDDT

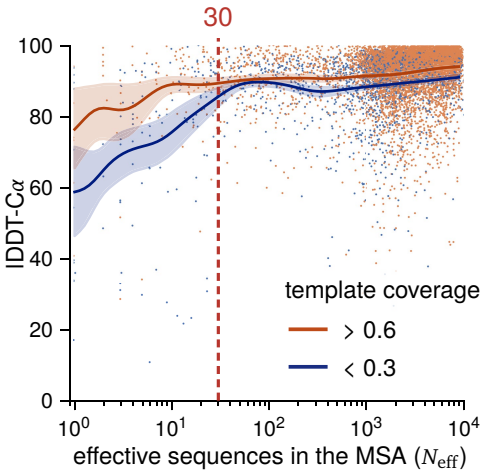



pLDDT is the predicted lDDT-C$\alpha$ per residue

**Predictions need a well-calibrated confidence**

K. Tunyasuvunakool et al., *Nature* 596, 2021.

# Limitations: MSA depth and complexes




Accuracy falls below about 30 effective sequences

AlphaFold is weaker for chains shaped by other chains

**AlphaFold needs enough related sequences**

# After AlphaFold 2: 2021 to 2025

| | |
|---|---|
| 2021 | RoseTTAFold appears online the same day |
| | 98.5% of human proteins are predicted |
| | AlphaFold-Multimer predicts complexes |
| 2022 | ESMFold uses a language model and no MSA |
| 2024 | AlphaFold 3 is one model for proteins, DNA, RNA and ligands |
| | Baker, Hassabis and Jumper receive the Nobel Prize in Chemistry |
| 2025 | AlphaGenome predicts regulatory activity from a million bases of DNA |

address the limits

**AlphaFold 3 is more accurate than specialised tools, with one exception**

The first authors are Baek 2021, Tunyasuvunakool 2021, Evans 2021, Lin 2022, Abramson 2024 and Avsec 2025.

# AlphaFold 3: which choices survived

**Abramson et al., *Nature* 630 (2024), pages 493 to 497**

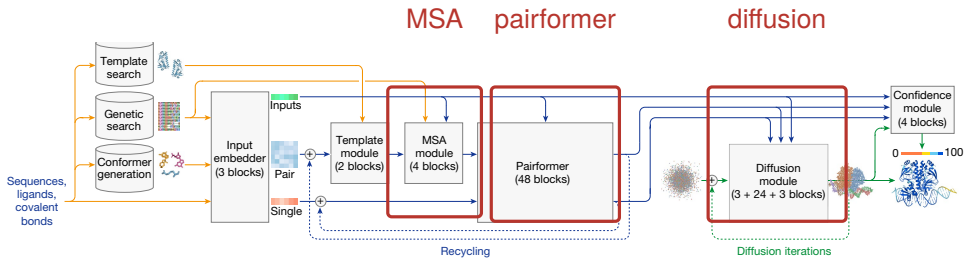
| | AlphaFold 2 | AlphaFold 3 |
|---|---|---|
| 1 | attention over sequences | a small MSA block, and the MSA is not kept |
| 2 | triangles on the pairs | kept, 48 blocks |
| 3 | frames, IPA, equivariance | dropped for diffusion and random rotations |
| 4 | FAPE in local frames | dropped, 4.4 % chirality violations remain |

**The bitter lesson, with two footnotes. The graph survived. The hand came back.**

Richard Sutton, *The Bitter Lesson*, 2019: “building in how we think we think does not work in the long run”.

# The AlphaFold 3 architecture

Abramson et al., *Nature* 630 (2024), Fig. 1d




Tokens for proteins, nucleic acids, ligands and ions

Diffusion denoises random atom coordinates, 25 samples ranked by confidence

**Less structure in the network, more in the data and in the sampling**

# AlphaFold 3: the cost of the removal

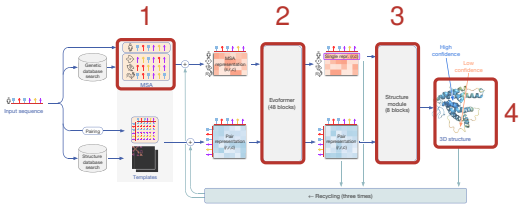
**Abramson et al., *Nature* 630 (2024), pages 495 to 498**

| AlphaFold 2 had | AlphaFold 3 reports | The patch |
|---|---|---|
| the hand in the loss | 4.4 % chirality violations | ranking, more seeds |
| a violation loss | clashes, overlapping chains | ranking |
| loops for disorder | hallucinated structure | AlphaFold-Multimer distillation |
| MSA attention | same MSA dependence | none |

No ablation of the removed parts is reported

**It wins everywhere but one RNA entry. The costs were paid in patches.**

# Summary: four design choices, measured




| | | GDT lost |
|---|---|---|
| 1 | attention over sequences | −2.2 |
| 2 | triangles on the pairs | −4.6 |
| 3 | frames and IPA | −2.3 |
| 4 | FAPE in local frames | mirrored |

Evolution and geometry are trained end to end, in about one GPU minute per model

**The median backbone error is below one atom width**

J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* 596, 583–589 (2021). GDT lost on CASP14 when removed, Fig. 4a and Suppl. Fig. 9.

# Exam question

Placeholder. The question from the course team arrives about seven days before the talk.

Our answer.

# Questions for the audience

**1** AlphaFold 2 built rotation equivariance and the hand into the network and the loss. AlphaFold 3 dropped both and learns them from data. When is it worth building a symmetry into a model, and when is it better learned?

**2** Accuracy falls below about 30 related sequences. Has the network learned how proteins fold, or how evolution has varied them? What result would settle it?

**3** pLDDT is trained on the model's own errors. On which proteins would you distrust it, and what should a genomics pipeline do with a low-confidence region?

# Why is IPA invariant to rigid motions?

SI 1.8.2, Algorithm 22, equations 3 to 6




The only geometry in the weights is the distance between points that $i$ and $j$ place in their own frames

$$\left\|(T_{\text{global}} \circ T_i) \circ \vec{\boldsymbol{q}}_i - (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{k}}_j\right\|^2$$
$$= \left\|T_{\text{global}} \circ (T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j)\right\|^2 = \left\|T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j\right\|^2$$

moving both points together leaves their distance unchanged

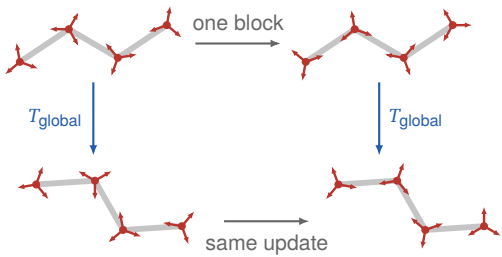
$$(T_{\text{global}} \circ T_i)^{-1} \circ \textstyle\sum_j a_{ij}\, (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{v}}_j$$
$$= T_i^{-1} \circ T_{\text{global}}^{-1} \circ T_{\text{global}} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j = T_i^{-1} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j$$

the output points are read back in frame $i$, and $T_{\text{global}}$ meets its inverse. Exact, not learned.

**Weights and outputs do not depend on where the structure sits**

# Why is the structure module equivariant?

SI 1.8.2 and 1.8.3, Algorithms 20 and 23




The update of a frame is predicted from $\boldsymbol{s}_i$, which is invariant, and composed in the frame's own coordinates

$(T_{\text{global}} \circ T_i) \circ T_i^{\text{upd}} = T_{\text{global}} \circ (T_i \circ T_i^{\text{upd}})$

moving the input first or the output last gives the same frames

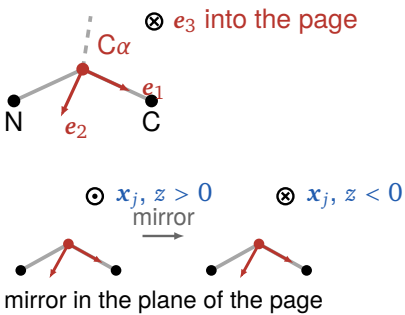
"Rigid motion" in the supplement means proper rotation and translation, not reflection. In group language: SE(3).

All frames start as the identity, so the output's global frame is arbitrary and the loss must be invariant to it

**Exact SE(3) equivariance. Reflections are excluded on purpose.**

# Where does the handedness come from?

SI 1.8.1, 1.9.3 and 1.8.4, Algorithm 21, equations 11 to 17




A frame is built from N, Cα and C by Gram–Schmidt, with $\boldsymbol{e}_3 = \boldsymbol{e}_1 \times \boldsymbol{e}_2$. It is always right-handed.

Reflect all atoms. The frames do not reflect with them. In local coordinates exactly $z$ flips:

$$T_i^{-1} \circ \boldsymbol{x}_j \;\mapsto\; \mathrm{diag}(1, 1, -1)\,(T_i^{-1} \circ \boldsymbol{x}_j)$$

$$\mathrm{FAPE}(X, X_{\mathrm{mirror}}) = 2 \sum_{ij} \left|(T_i^{-1} \circ \boldsymbol{x}_j)_z\right|$$

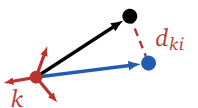
large unless the structure is flat

Trained with dRMSD, the error of the distance matrix, which is blind to reflection, the model builds the mirror image in a large share of cases (Suppl. Fig. 9)

**The loss sees the hand, the data teach it. Side chains are always built L.**

# What does FAPE measure?

SI 1.9.2, 1.9.4 and 1.9.5, Algorithm 28, equations 18 to 30




seen from frame $k$

Every atom $i$ in every frame $k$, in local coordinates: $N_{\text{frames}} \times N_{\text{atoms}}$ terms, each clamped at 10 Å and divided by 10 Å

Invariant to a rigid motion of the prediction, and of the truth, by the same cancellation as in IPA

Lower bound: FAPE is at least the RMSD after the best rigid alignment, whatever the frames (eqs. 18 to 22). So it is zero only if the two point sets differ by a proper rigid motion.

Non-negative, symmetric, triangle inequality: a pseudometric on structures (eqs. 23 to 30)

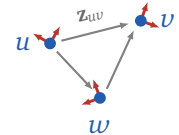
**Local geometry in every frame: blind to placement, sensitive to the hand**

# Is AlphaFold 2 a graph neural network?

**Formalism of Bronstein, Bruna, Cohen and Veličković, 2021**

$$\mathbf{h}_u = \phi\Big(\mathbf{x}_u, \bigoplus_{v \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv})\Big)$$

message $\psi$, aggregation $\bigoplus$, update $\phi$. Nodes: residues, $\mathbf{s}_u$ and frame $T_u$. Edges: $\mathbf{z}_{uv}$.




| Block | message $\psi$ | into |
|---|---|---|
| MSA row attention | $\psi(\mathbf{m}_{su}, \mathbf{m}_{sv}, \mathbf{z}_{uv})$, the edge biases the weight | node $u$ |
| outer product mean | $\psi(\mathbf{m}_{\cdot u}, \mathbf{m}_{\cdot v})$, averaged over the sequences | edge $uv$ |
| triangle updates | $\psi(\mathbf{z}_{uw}, \mathbf{z}_{wv})$ over every third node $w$ | edge $uv$ |
| IPA | $\psi(\mathbf{s}_u, \mathbf{s}_v, \mathbf{z}_{uv}, \lVert T_u \circ \vec{\mathbf{q}}_u - T_v \circ \vec{\mathbf{k}}_v \rVert)$, values moved by $T_u^{-1} \circ T_v$ | node $u$ |
| backbone update | none, $\phi$ alone | frame $T_u$ |

Triangles are two-hop messages. They make the edges consistent with one structure.

**Geometry enters only through distances. A frame is a gauge fixed by its atoms.**

# What does each block compute?

## Supplement, Algorithms 6 to 32

| | Update | Message | SI |
|---|---|---|---|
| Evoformer, 48 blocks | | | |
| 1 | $\mathbf{m} \leftarrow \mathbf{m} + \text{RowAttention}(\mathbf{m} \mid \mathbf{z})$ | node → node, edge biases the weight | Alg. 7 |
| 2 | $\mathbf{m} \leftarrow \mathbf{m} + \text{ColumnAttention}(\mathbf{m})$ | across the sequences at one node | Alg. 8 |
| 3 | $\mathbf{m} \leftarrow \mathbf{m} + \text{MLP}(\mathbf{m})$ | per node | Alg. 9 |
| 4 | $\mathbf{z} \leftarrow \mathbf{z} + \text{OuterProductMean}(\mathbf{m})$ | node pair → edge | Alg. 10 |
| 5 | $\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleMultiplication}(\mathbf{z})$, out and in | edge pair → edge | Alg. 11, 12 |
| 6 | $\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleAttention}(\mathbf{z})$, start and end node | edge pair → edge | Alg. 13, 14 |
| 7 | $\mathbf{z} \leftarrow \mathbf{z} + \text{MLP}(\mathbf{z})$ | per edge | Alg. 15 |
| Structure module, 8 blocks with shared weights. At the start $\mathbf{s}_u = \mathbf{m}_{1u}$ and $T_u$ is the identity. | | | |
| 8 | $\mathbf{s} \leftarrow \mathbf{s} + \text{IPA}(\mathbf{s}, \mathbf{z}, T)$ | node → node, values moved by $T_u^{-1} \circ T_v$ | Alg. 22 |
| 9 | $\mathbf{s} \leftarrow \mathbf{s} + \text{MLP}(\mathbf{s})$ | per node | Alg. 20 |
| 10 | $T_u \leftarrow T_u \circ \text{BackboneUpdate}(\mathbf{s}_u)$ | per node, in its own frame | Alg. 23 |
| Recycling, 4 cycles | | | |
| 11 | $\mathbf{z} \leftarrow \mathbf{z} + \text{Embed}(d_{uv})$, C$\beta$ distances from the frames | frames → edges | Alg. 32 |

Every line is $\mathbf{h}_u = \phi(\mathbf{x}_u, \bigoplus_v \psi)$ on nodes or on edges. Information moves in lines 1, 4, 5, 6, 8 and 11.

# What exactly did each ablation remove?

SI 1.13.1 to 1.13.3

| Row of Fig. 4a | What the model has instead |
|---|---|
| No raw MSA | the profile and the pairwise frequencies of the MSA, in one row, no column attention |
| No triangles, biasing or gating | axial attention on the pairs with a relative position bias, no gating |
| No IPA | frames from one linear projection of the single representation, no iteration |
| No recycling | a single pass through the network |
| No end-to-end structure gradients | the structure losses stop before the Evoformer, which learns from the auxiliary losses only |
| No templates | no template stack, four more MSA clusters instead |

Baseline: the full model without self-distillation. Three or four seeds, not re-tuned

**Not additive: no IPA costs 2.3, no recycling 4.5, both together 16.3 GDT**

# How is confidence predicted?

**SI 1.9.6 and 1.9.7, Algorithm 29**

pLDDT: a small network on the final single representation predicts, per residue, a distribution over 50 bins of lDDT-C$\alpha$. It is trained with cross-entropy against the lDDT-C$\alpha$ of the prediction itself, on structures resolved between 0.1 and 3 Å

The reported pLDDT is the expected value of that distribution. The pLDDT of a chain is the mean over its residues

pTM, the predicted template modelling score: a pairwise head predicts the distribution of the aligned error of every residue pair, and the TM-score follows from these distributions

**Confidence is a prediction, trained on the model's own errors**

# How was it trained?

**Methods, “Training regimen”, SI 1.3 and 1.11.1, Table 4**

Data: PDB structures released before 30 April 2018, and about 350,000 predicted structures of Uniclust30 sequences without a structure, filtered by confidence, 75 % of the samples

Stage 1: crops of 256 residues, about 10 million samples, about seven days on 128 TPU v3 cores

Stage 2: crops of 384 residues, the structural violation loss switched on, about 1.5 million samples, about four days

Losses: FAPE, torsion angles, distogram, masked MSA, pLDDT, experimentally resolved, violations

**Eleven days on 128 TPU cores per model. Inference takes GPU minutes to hours.**

# AlphaFold 3: further changes

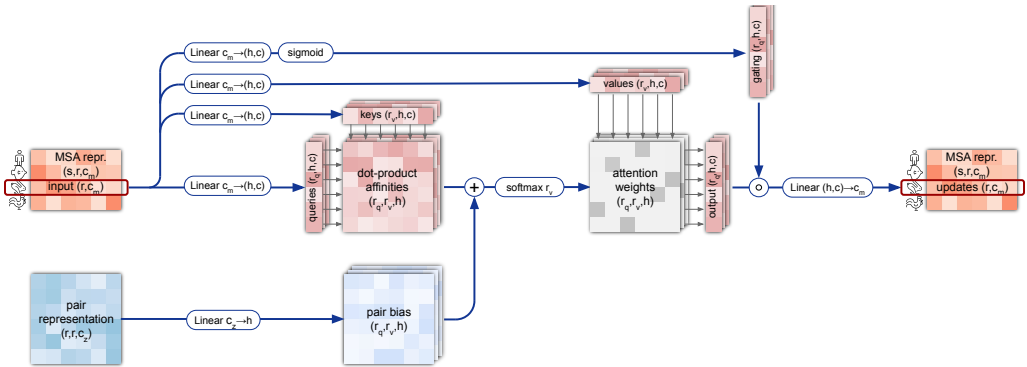
Abramson et al., *Nature* 630 (2024)

| | | |
|---|---|---|
| Input | tokens, one per residue or per ligand atom | p. 493 |
| Trunk | pairformer, 48 blocks. MSA block of 4, MSA not kept | p. 494 |
| Output | diffusion over raw atoms, no frames, random rotations instead | p. 495, SI Alg. 19 |
| Confidence | trained on 20-step mini rollouts | p. 495 |
| Sampling | 5 seeds, 5 samples each, ranked with clash and chirality penalties | p. 494, 497 |
| Limits | chirality 4.4 %, clashes, hallucination, no dynamics | p. 496, 497 |

**Less structure in the network, more in the data and in the sampling**

# MSA row attention with pair bias

## Supplementary Fig. 2, Algorithm 7




**Supplementary Fig. 2. Every sequence of the MSA attends along itself, the pair representation biases the weights**